

Smart Lab Equipment & slot Reservation portal

Requirements Table

ID	Type	Description	priority	Acceptance Criteria	Rationale	comments
FR-001	Functional	The system shall allow authenticated students to reserve calibrated equipment for a maximum of 2 hours, up to 7 days in advance.	High	Pass if valid reservations meet both limits and equipment is available	Controls reservation duration and advance booking.	Verify that the 2-hour and 7-day limits are enforced.
FR-002	Functional	The system shall allow reservations only for equipment with valid calibration.	High	Pass if uncalibrated equipment cannot be reserved.	Ensures safe equipment usage.	Calibration status must be checked before reservation.
FR-003	Functional	The system shall reject reservations that overlap an existing reservation for the same equipment.	High	Pass if every overlapping request is rejected.	Prevents double-booking.	Test simultaneous and overlapping reservation requests.
FR-004	Functional	The system shall allow students to cancel future reservations before their start time.	Medium	Pass if the reservation is cancelled and the slot becomes available.	Releases unused equipment slots.	Specify that cancellation is allowed only before the start time.
FR-005	Functional	The system shall record the equipment return time and identify late returns.	High	Pass if return time > scheduled end time is marked Late.	Supports late-return rules.	Confirm that actual and scheduled return times are recorded.
NFR-001	Non-Functional	The system shall process concurrent reservation requests within 200ms using database-level locking.	High	Pass if requests complete within 200ms without double-booking.	Prevents race conditions.	Test under concurrent reservation requests.
NFR-002	Non-Functional	The system shall require authentication before reservation operations.	High	Pass if unauthenticated users cannot create or cancel reservations.	Prevents unauthorized access.	Test access with an unauthenticated user.